

SAT 3812: Cyber Security I

Instructor: *Yunlong Xing*

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Who am I?

- Dr. Yunlong Xing (*Pronounced: Yoon-loong Shing*)
 - Ph.D. in Information Technology at George Mason University
 - Research Interests: Software and Systems Security
 - Personal Website: <https://yunlongxing.github.io>
 - Email: yuxing@mtu.edu
 - Office: EERC 220
- New faculty at Michigan Technological University
 - Any suggestion is welcom

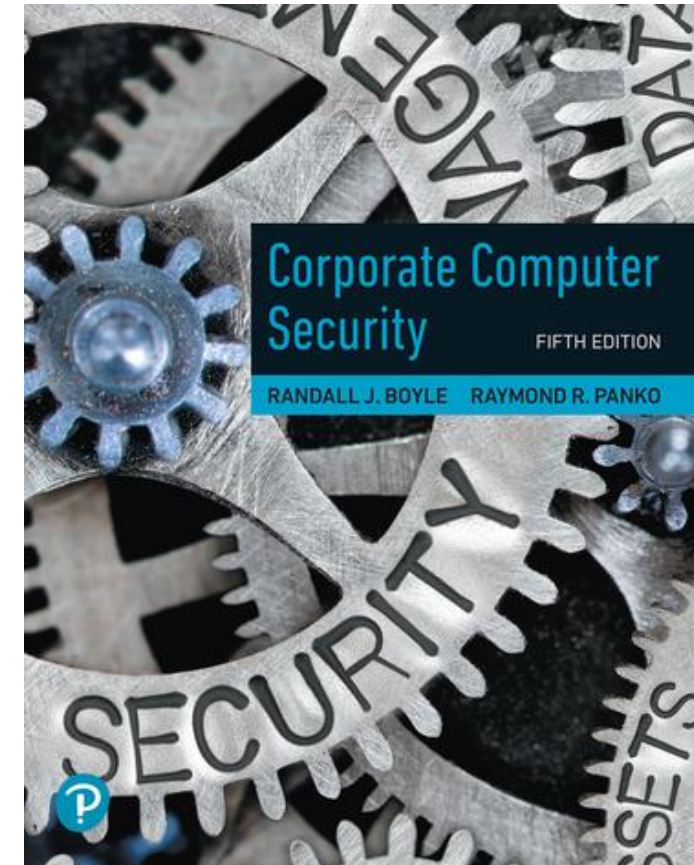
Teaching Philosophy

- Bridge the gap between societal demands and the readiness of undergraduate students.
- Blend practical real-world exposure with academic requirements.
- Aim to cultivate a rich and captivating learning atmosphere.
- Instill in students not just the 'what' of knowledge, but also the 'how' of learning.
- Cultivate self-directed learning capabilities.

Course Introduction

■ Syllabus

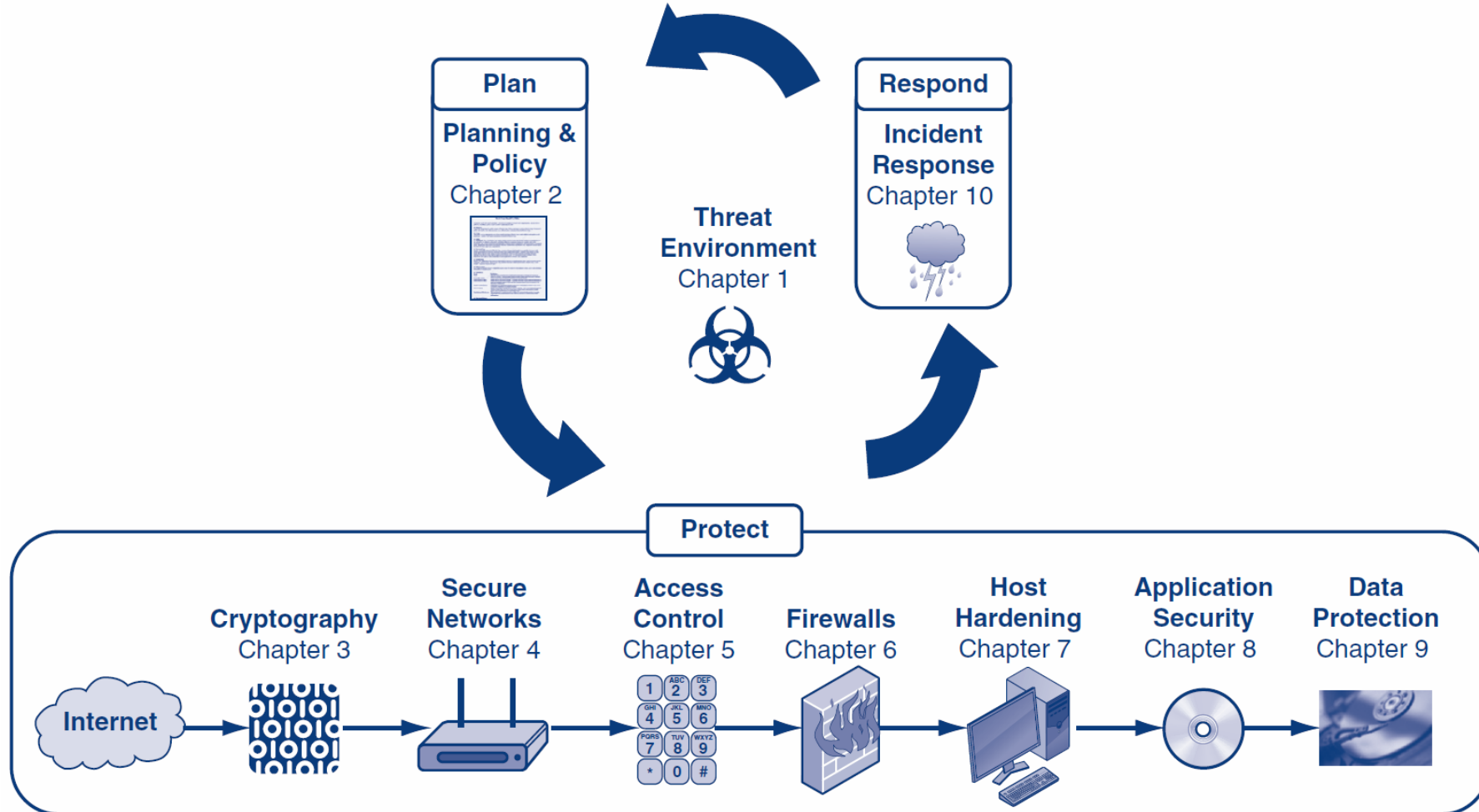
- Textbook (5th edition)
- Lecture / Lab arrangement
- Attendance
- Office hour
- Grading policy
- Mistakes / Integrity
- Canvas page



Entry Survey

- This is a survey to gauge your background knowledge in Linux and cybersecurity. This survey is for informational purposes and will not contribute to your final course grade.
- You should aim to get a score of 6 or more (out of 10). This shows you have necessary background knowledge for the course. If you don't, you should consider to take the course at a later time or discuss your other options with your academic / faculty advisors.
- Open book, open notes, open Internet access, work independently. Do not use ChatGPT.

How is the textbook organized?



Textbook & Course

- This is a comprehensive textbook on network security, covering almost all important topics.
- This is a book about security defense, not how to attack (offensive security).
- We will use the textbook, by adding a lot of case studies and other materials to teach you how hackers attack.

We Learn Together!

- We aim to create a welcoming and inclusive learning environment in this course.
 - Certain information discussed may be delicate. Please don't interpret anything as offensive. Should you have reservations about any subject within this course, please don't hesitate to approach me during or after class for a discussion.
- Abusing technology such as hacking is absolutely forbidden within and outside of this course.
 - Our curriculum involves understanding hacker tactics to enhance your defensive skills. It is imperative that you refrain from any hacking activities. Engaging in such activities could lead to serious consequences that you undoubtedly want to avoid.

Honor Code

- Zero-Tolerance on plagiarism
 - Students are required to follow the university guidelines on academic conduct at all times. Note that students are explicitly forbidden from copying anything off of the Internet (e.g., source code, text, slides) **without proper acknowledgement**, using anything from an answer guide, copying code/answers from each other to complete any assignment or a course project.
- Usage of ChatGPT and other AI tools are allowed in this course, with proper disclosure and acknowledgement.

What are we going to learn in this course?

- The basic concepts of computer and network security
- Planning & policy
- Cryptography
- General network security
- Access control
- Firewall
- Host hardening
- General application security
- Data security
- Disaster response

College Virtual Machine Platform

- Virtual Server Cluster
 - Over \$250k investment
 - Managed by IT
 - Fast, reliable, and convenient
 - Heavily used by CNSA/IT, Cybersecurity, CS, and ECE students
 - The industry trend
 - Cloud and virtualization

Web, Linux, Python, and Network Primer

- Web Primer

- HTML/CSS/Javascript and PHP for web programming
- Basic understanding of web programming

- Linux Primer

- Short version and long version
- Get familiar with commonly-used Linux commands

- Python Primer

- Basic python programming

- Network Primer

- Basic TCP/IP and networking concepts

Week	Monday	Wednesday	Notes
Week 1	Course introduction; Lecture 1 – Overview	Lecture 2 – Policy	No Lab; Entry Survey
Week 2	LABOR DAY, No class	Lecture 2 – Policy (cont.) – Watch Video. K-day (noon Friday)	Lab 1; Quiz 1
Week 3	Lecture 2 – Policy (cont.) Target Case Study	Lecture 3 – Email Security	Lab 2; Quiz 2; Risk Analysis
Week 4	Lecture 4 – Web Intro	Lecture 5 – Web Security Career Fair (Tues 2-4pm)	Lab 3; Quiz 3; Early-term Survey
Week 5	Lecture 5 – Web Security	Lecture 5 – Web Security (cont.)	Quiz 4; Bonus: AI Coding
Week 6	Lecture 6 – Network Security	Lecture 7 – DDoS	Lab 4; Quiz 5
Week 7	Lecture 7 – DDoS (cont.) Mid-term Review	Lecture 7 – DDoS 2	Lab 5
Week 8	Mid-term Exam: Take Home	No class	No Lab; No Quiz
October Recess (Thur/Fri); Spring/Summer Registration Begins			
Week 9	Lecture 8 – Access Control	Lecture 8 – Access Control (cont.)	Lab 6; Quiz 6
Week 10	Lecture 9 – Firewall	Lecture 9 – Firewall (cont.)	Lab 7; Quiz 7
Week 11	Lecture 10 – Malware	Lecture 10 – Malware (cont.)	Lab 8; Quiz 8
Week 12	Lecture 11 – Cryptography	Lecture 11 – Cryptography (cont.)	Course Project
THANKSGIVING BREAK			
Week 13	Lecture 11 – Cryptography (cont.)	Lecture 11 – Cryptography (cont.)	Lab Final Exam
Week 14	Review and Presentation	Review and Presentation	No Lab; No Quiz
Final Exam	Final exam as scheduled by Registrar's office, schedule TBD. 12/19: Midyear Commencement		

Cybersecurity Certifications

- Information on Cybersecurity related Certifications
 - Security +
 - CISSP: SAT5283
 - Certified Ethical Hackers
 - Will have a separate talk on this

RedTeam

- RedTeam: an on-campus student organization focused on cybersecurity
 - Over 30 student members; meets every week during the semester
 - Various RedTeam activities such as seminars, competitions, and conference visits.
 - Regular attenders of the National Cyber League (NCL) competition
 - Dr. Yu Cai and Dr. Bo Chen are the faculty advisors of the RedTeam

A Taste of Cybersecurity

What is Cybersecurity?

- An academic Definition:
 - The protection of computer and cyber systems from theft, damage and misuse to their hardware, software, data, and services.
 - Cybersecurity is a very broad term...
- A simpler definition:
 - Safeguarding the Cyber World

Why Cybersecurity is Important?

Physical world vs Cyber world

- The world has increasingly become a combination of the physical world and the cyber (digital) world.
- Example of cyber world:
 - Facebook....
 - Online bank...
 - Games...

Why Cybersecurity is Important?

Physical world vs Cyber world

- As human beings, we lived in the physical world for thousands of years.
- The cyber world is relatively new. Many people don't have a good idea on how to protect themselves.
 - You are the **Cyber Ambassadors!**
 - Protect yourself and teach others

What is Ethical Hacking

- Hacking is illegal
- Ethical Hacking is an **authorized** act of penetrating into systems to find out threats and vulnerabilities.
- The purpose is to **improve security** by fixing the vulnerabilities found during testing.
- Also called: Penetration Test

Who are the hackers?



Hackers (for profit or fame)

Organized Crime (for profit)



Nation States & Competitors (Espionage and Cyberwarfare)



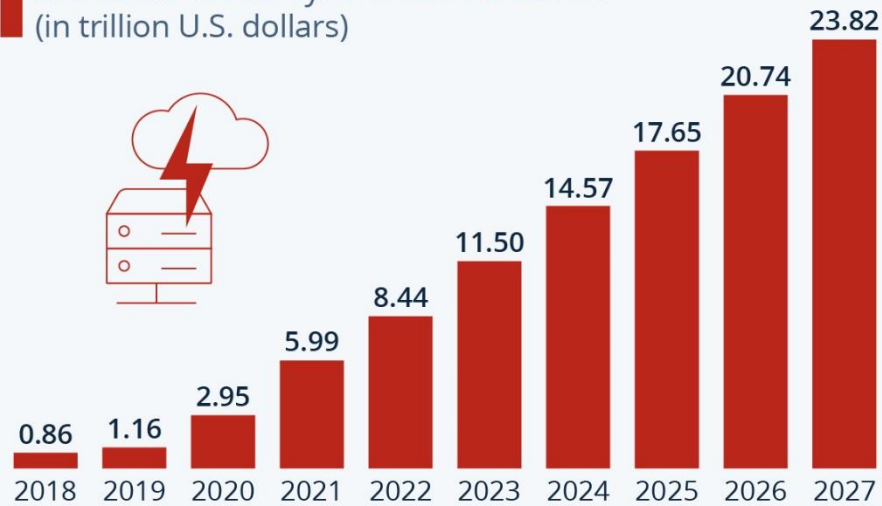
Hacktivists (for a political purpose, agenda, or belief system)



Researchers (Techies, students, script kiddies)

Cybercrime Expected To Skyrocket in the Coming Years

Estimated cost of cybercrime worldwide
(in trillion U.S. dollars)



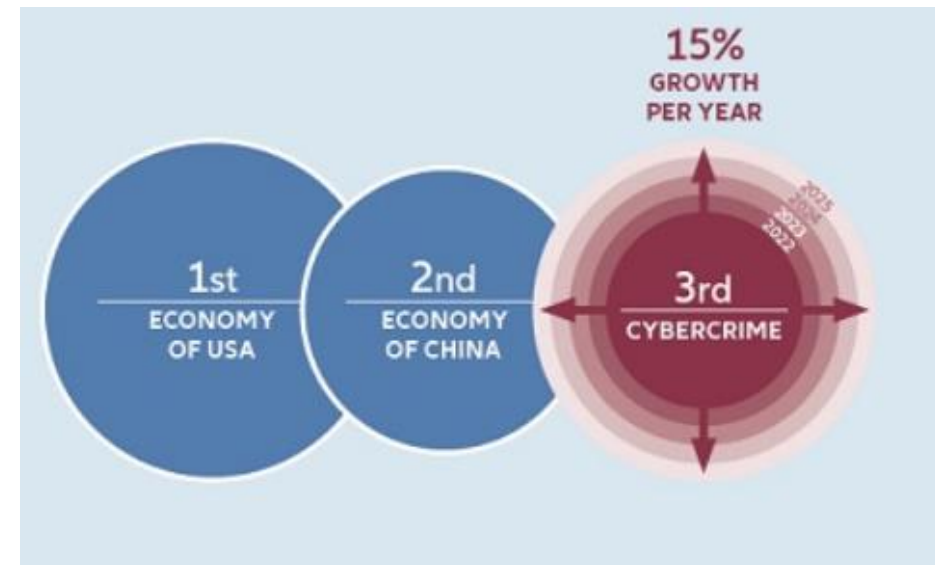
As of November 2022. Data shown is using current exchange rates.

Sources: Statista Technology Market Outlook,
National Cyber Security Organizations, FBI, IMF



statista

Cybercrime is one of the largest and fastest-growing enterprises in the world

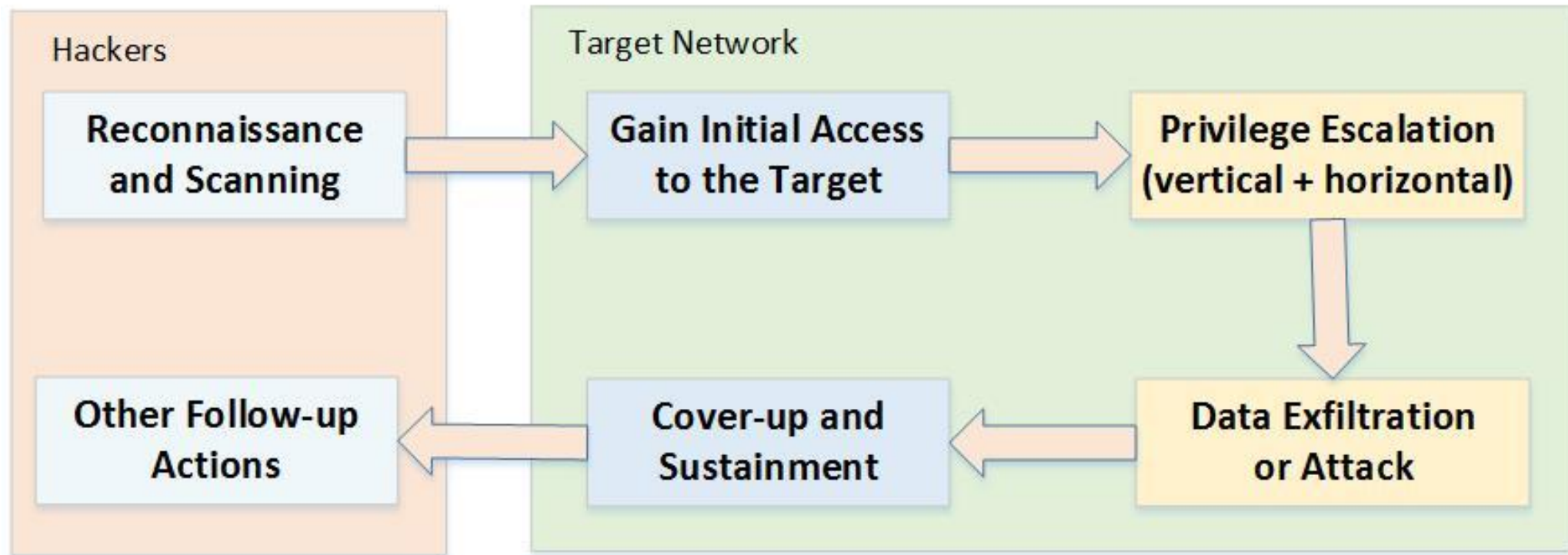


Top areas where security leaders are increasing their investment



Source: Team8: 2021 Cybersecurity Brief, as of 26 January 2021

Typical Steps in Cyber Attacks



Demo: Your password is for sale!

- <https://haveibeenpwned.com/>
 - A website which allows you to get password leaking information.
 - Exercise
 - Search your own email address and see if your accounts have been compromised!

Demo: Make Good Use of Google Search

- Exercise
 - Please use Google Search to find out who is the governor of Michigan? Please find out the governor's email address.

Demo: Get Bulk Email Addresses For Free

- You can get bulk email addresses from <http://hunter.io>
- You may need to register an account.
- Demo: Search for mtu.edu domain.
- Exercise: Search for Michigan.gov domain. Find a few email addresses from Michigan.gov.

Exercise : Cybersecurity Job Market

- Visit: <https://www.cyberseek.org/>
 - Study the interactive heat map of cybersecurity supply and demand.
 - Question: which state has the most cybersecurity job openings?
 - Study the Cybersecurity Career Pathway.

Some Recent Cyber Security Breaches

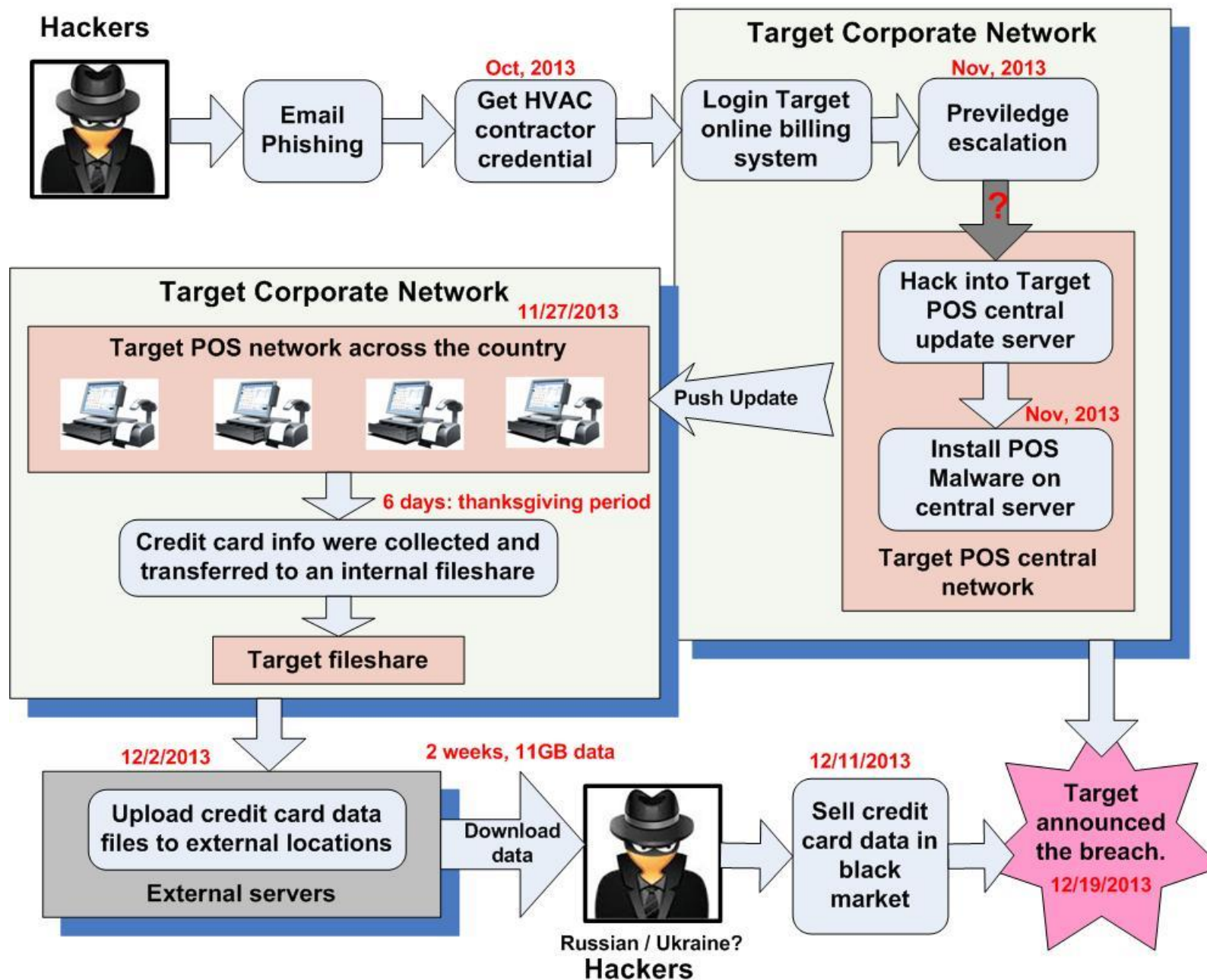
Colonial Pipeline Cyber Attack: 2021

- <https://www.youtube.com/watch?v=J1SXfM1yrvM> (3m)
- May 7, 2021, Colonial Pipeline suffered a **ransomware** cyberattack
 - Impacted computerized equipment managing the pipeline.
 - A regional emergency declaration for 17 states and D.C.
 - Colonial Pipeline paid the requested ransom (about \$4.4 million)
 - Recovered over 60% later by another group of hackers
 - **The largest cyber attack on U.S. infrastructure**

Target Data Breach in 2013

- In Dec. 2013, Target Corporation announced that data from around 70 million credit and debit cards was stolen.
 - <https://www.youtube.com/watch?v=w1o52wMzjFw>
- The breach started with phishing email attacks on a HVAC third-party contractor of Target.

The Target Breach 2013



Anthem Data Breach in 2015

- In Feb. 2015, Anthem suffered a data breach of nearly 80 million records, including personal information such as names, Social Security numbers, dates of birth, addresses, and other sensitive details
 - <https://www.youtube.com/watch?v=xyVKYHyCbKQ>
 - Thousands of fraudulent tax returns...
- Anthem breach also started with email phishing attacks.

Demo: Cyber Threat Real-time Map

Visit:

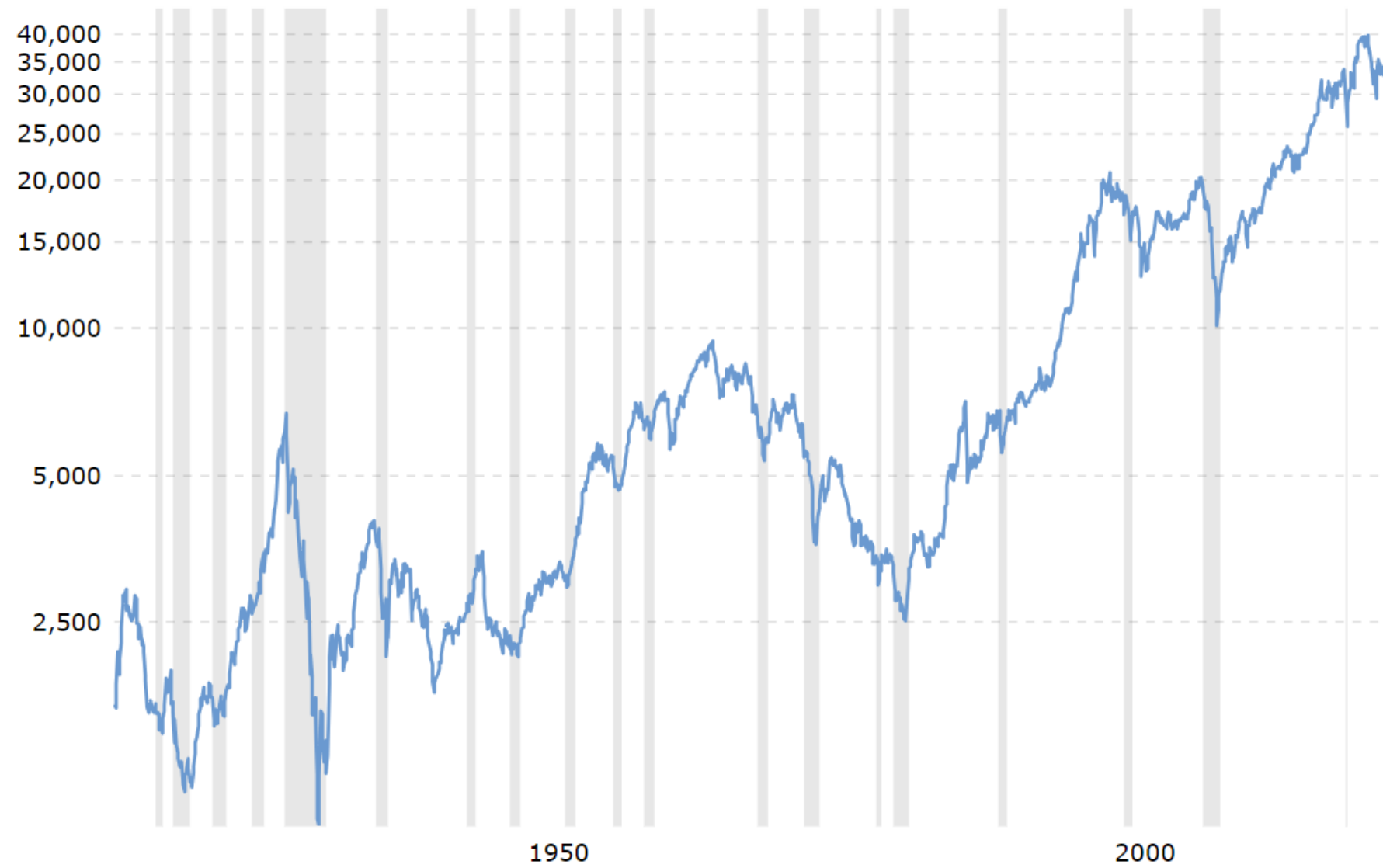
- <https://threatmap.bitdefender.com/>
 - Google keywords: threat map bitdefender
 - You can view attack time, attack type, attack country and target country

Exercise: World's Biggest Data Breaches & Hacks

Visit:

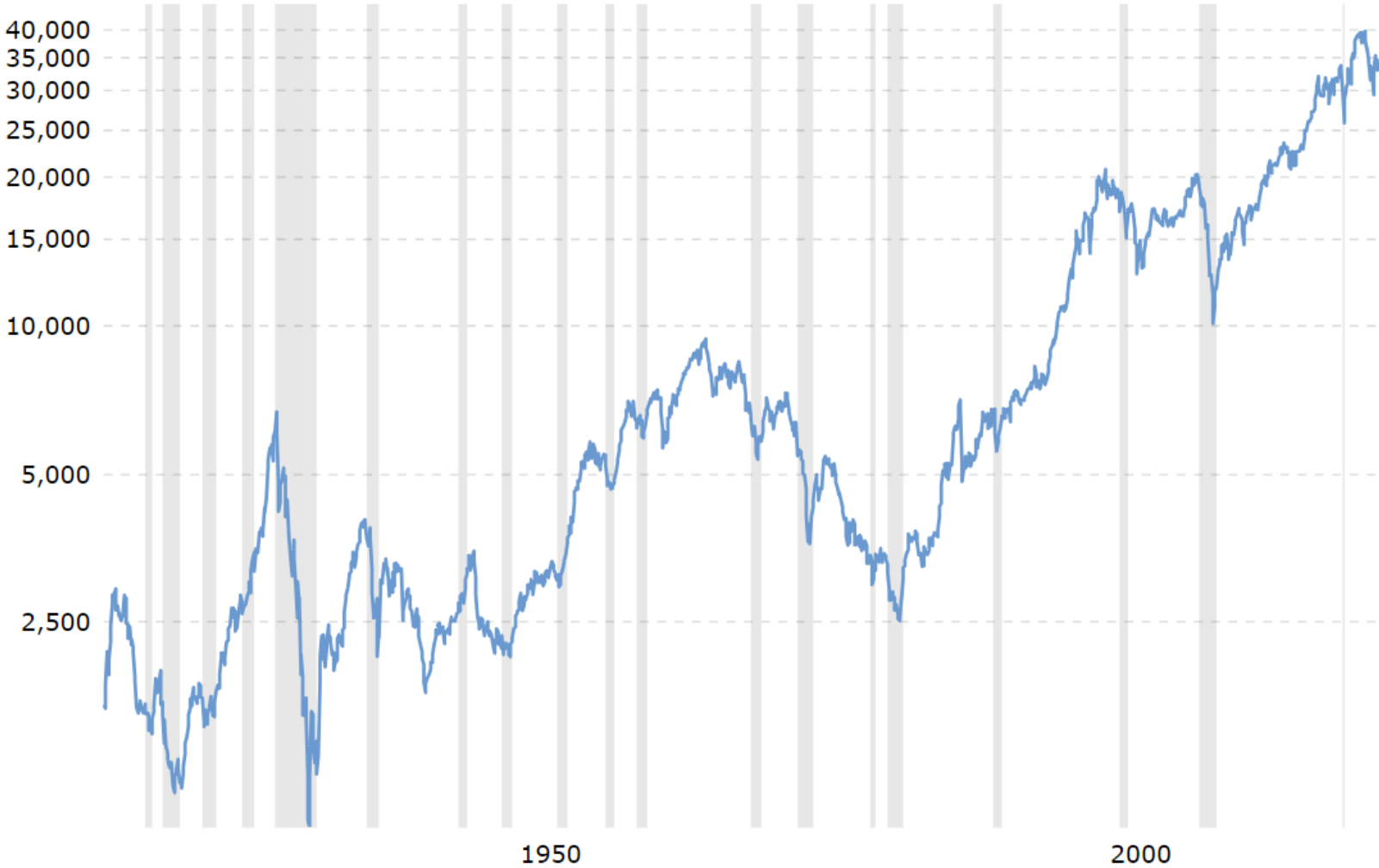
- <https://www.informationisbeautiful.net/visualizations/worlds-biggest-data-breaches-hacks/>
 - What is the biggest data breach you saw? (in term of number of users impacted)

Plan your career

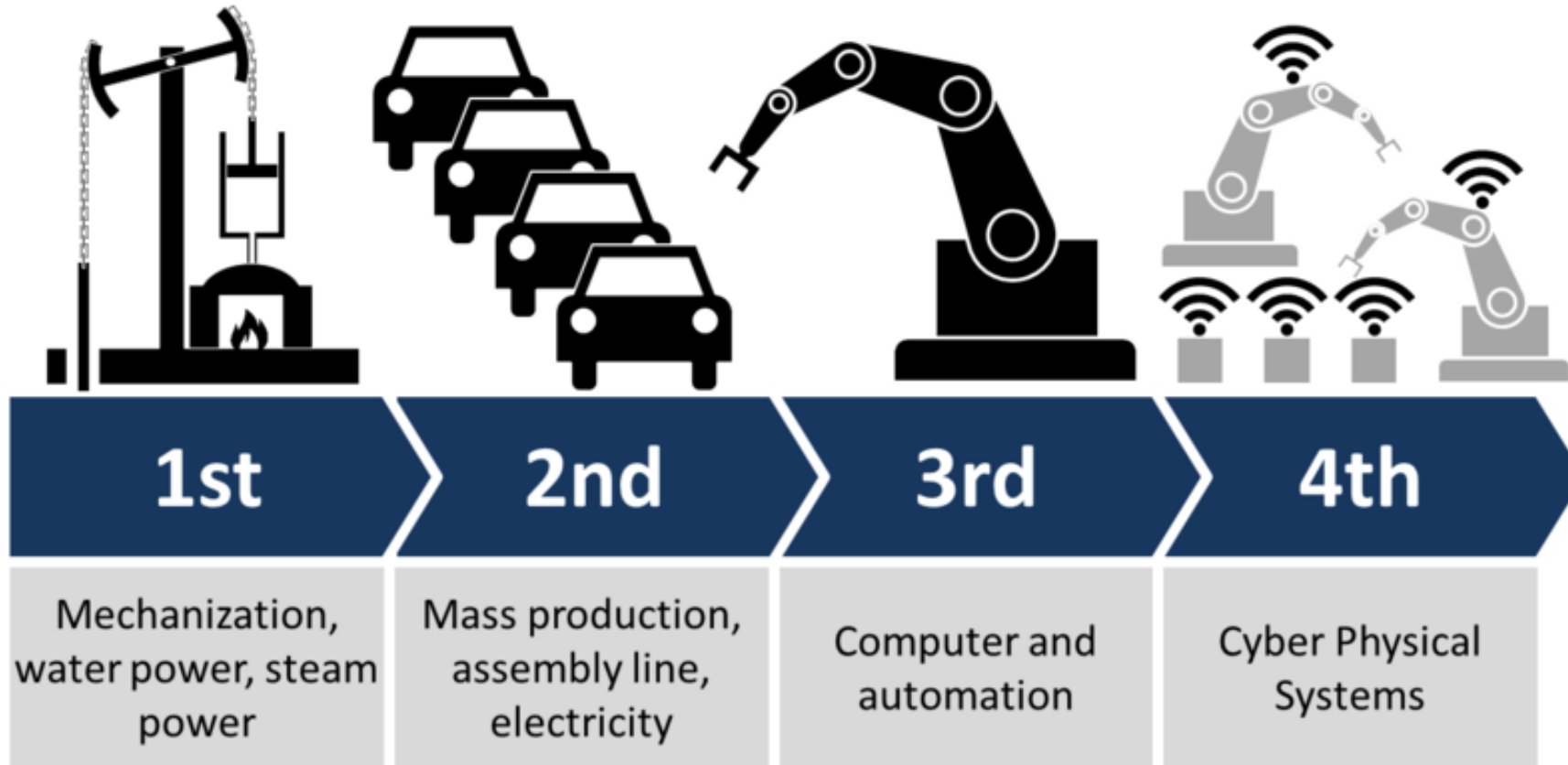


Dow Jones Index – 120 Year Chart

What does the history tell us?

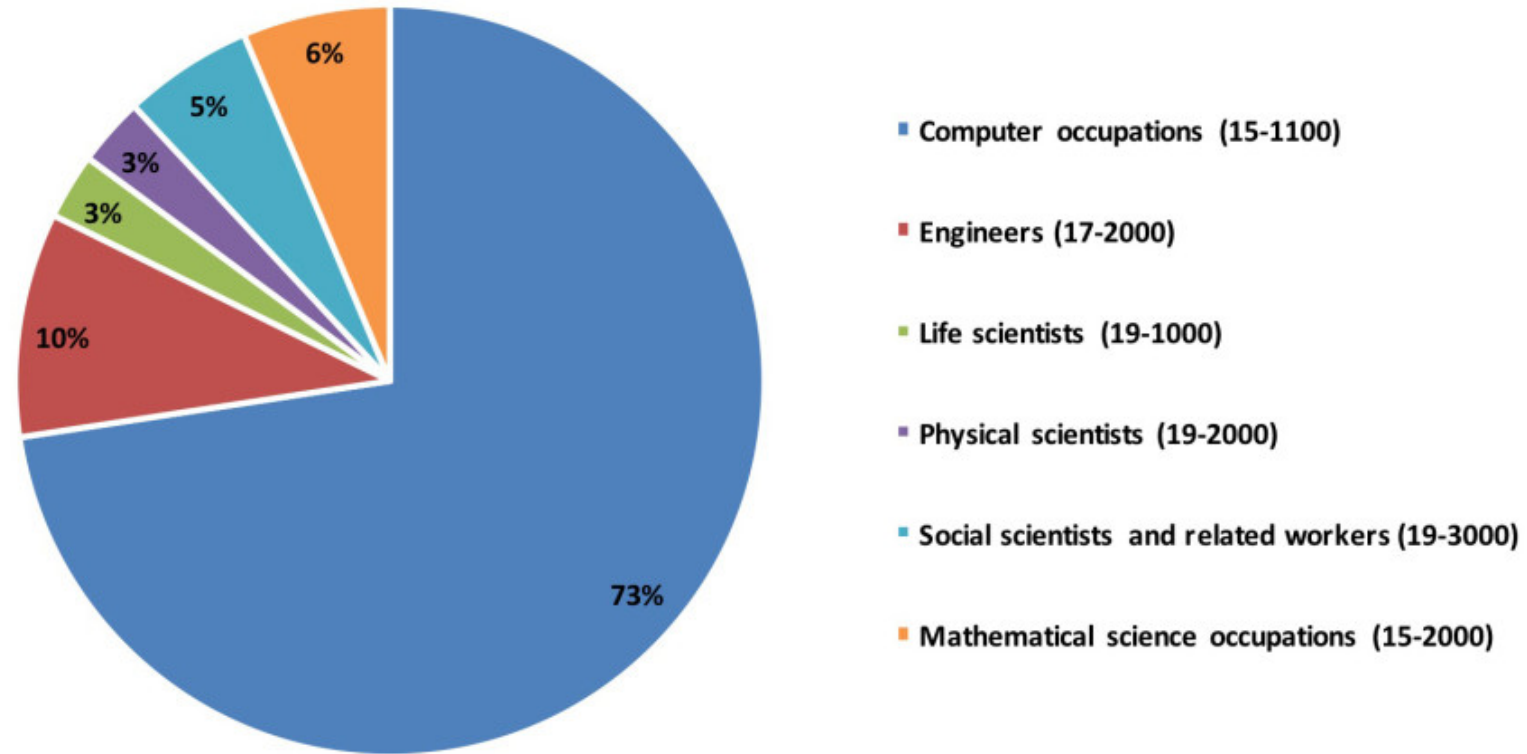


Four Industrial Revolutions



Where the Jobs Are

Job Growth, 2014-24 - U.S. Bureau of Labor Statistics



Data from the spreadsheet at <http://www.bls.gov/emp/ind-occ-matrix/occupation.xlsx>

Market Cap

2001 Market Cap

1. General Electric
2. Microsoft
3. Exxon Mobile
4. Wal-Mart
5. Citigroup



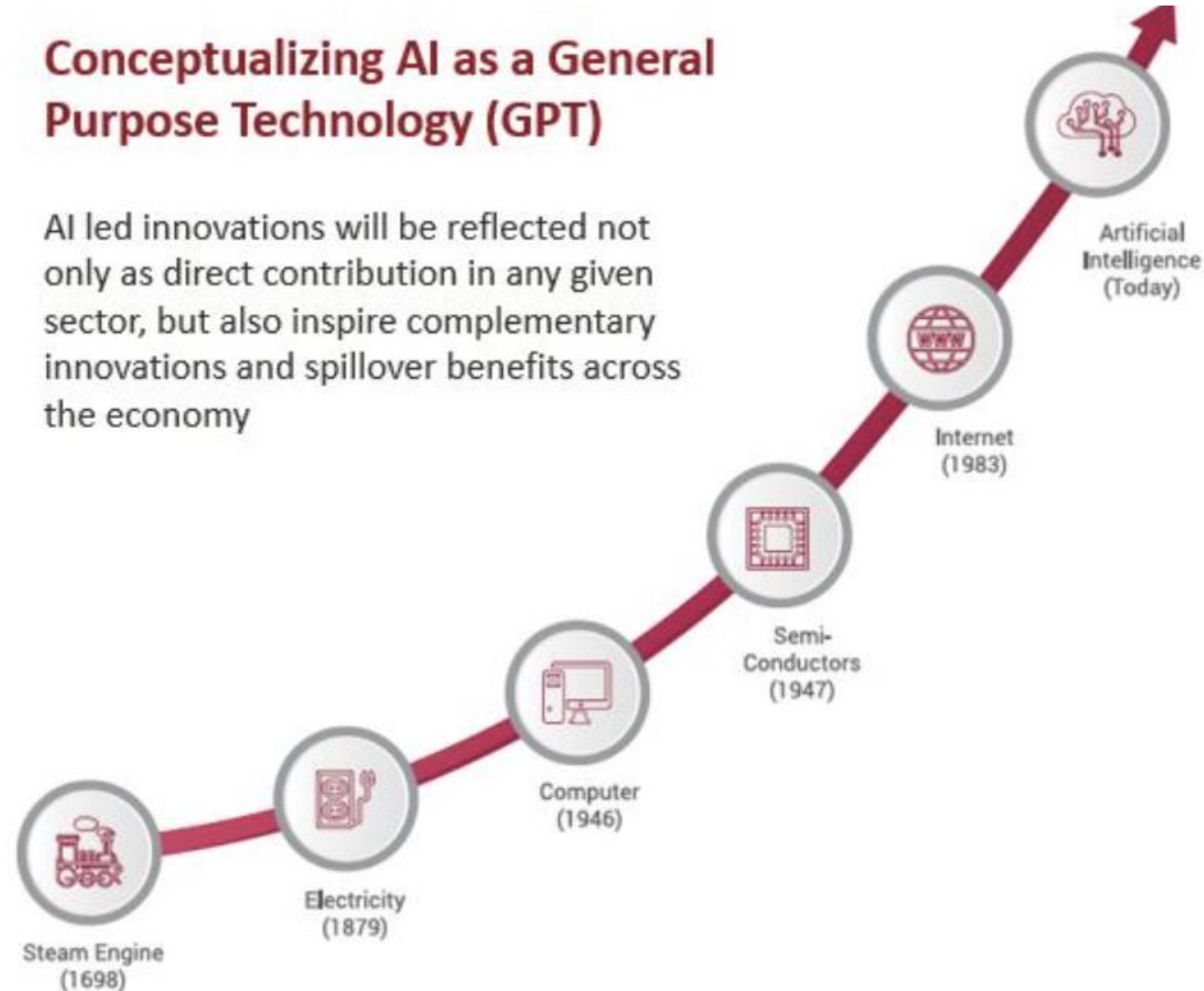
2023 Market Cap

1. Apple
2. Microsoft
3. Saudi Aramco
4. Alphabet, Inc.
5. Amazon

AI is going to change every aspect of our society within the next decade!

Conceptualizing AI as a General Purpose Technology (GPT)

AI led innovations will be reflected not only as direct contribution in any given sector, but also inspire complementary innovations and spillover benefits across the economy



Historical Instantiations of GPTs

One last piece of advice

“The only way to do great work is to love what you do.”

— Steve Jobs

If you haven't found it yet, keep looking. Don't settle.

STEVE JOBS Stanford Commencement Speech 2005 (15m)

<https://www.youtube.com/watch?v=sFaLN4xITTI>